

Valerie Boctor

☎ 714-345-1574 | ✉ valerieboctor@gmail.com | [in linkedin.com/in/valerie-boctor](https://www.linkedin.com/in/valerie-boctor) | [🌐 valerieboctor.github.io](https://valerieboctor.github.io)

Berkeley-trained Ph.D. economist specializing in consumer behavior and causal inference, with experience across tech, finance, and policy. I use large-scale data to understand how consumers make decisions and translate those insights into business and policy strategy.

EXPERIENCE

Research Data Scientist at Google, LLC

Mountain View, CA — Jun. 2026 - Present

- Conduct empirical research on consumer behavior, advertising measurement, and market insights using large-scale data and causal inference methods

Principal Quantitative Modeler at Capital One, NA

McLean, VA — Aug. 2025 - June 2026

- Developed default and loss forecasting models for the acquired Discover Personal Loan portfolio, valued at approximately \$10B, in collaboration with a team of STEM PhDs
- Built ETL pipelines to process loan performance data for 20 million U.S. personal loans
- Applied XGBoost, linear regression, and statistical segmentation methods to identify market segments critical to credit risk modeling
- Presented findings from a novel personal loans risk model to senior leadership, contributing to the team's Models and Data Groundbreakers Award in Q1 2026

Senior Associate at the Bank for International Settlements (BIS)

Basel, Switzerland | May 2021 – Jul. 2021

- Collaborated with BIS economists to produce timely research on the impact of fiscal deficits on inflation risks in the global economy during COVID, featured in the [IMF Seminars Recommended Reading](#).
- See Projects section below for additional details.

Compliance Risk Policy Analyst at the Office of the Comptroller of the Currency (OCC)

Washington, D.C. | June - Aug. 2017

- Designed logistic regression models to identify banks with heightened risk of compliance violations.
- Processed and summarized unstructured data on key compliance and consumer complaints for OCC bulletins.
- Developed technical risk assessments and policy briefs on compliance violations for the Acting Comptroller.

EDUCATION

Ph.D. in Economics

Dissertation: "Essays in Household Finance and Empirical Macroeconomics"

University of California, Berkeley

May 2025

B.S. in Economics, Political Science, Modern Foreign Languages

Summa cum laude, Phi Beta Kappa

Syracuse University

May 2018

PROJECTS

Mortgage Forbearance and Long Run Financial Distress

- Compiled data and constructed a panel dataset of 500,000 mortgaged homeowners from a 100TB SQL database containing consumer credit reports.
- Implemented an instrumental variables research design to show that mortgage forbearance significantly reduced mortgage delinquency and foreclosure with spillovers to revolving credit stability.
- Designed and estimated an event study model to compare financial outcomes for forbearance users with different relief durations and exit plans.

On Eliciting Subjective Probability Distributions of Expectations

with Olivier Coibion, Yuriy Gorodnichenko, & Michael Weber | NBER Working Paper

- Conducted exploratory regressions to analyze the impact of political uncertainty on economic behavior, using data from a randomized control trial of 20,000 American households.

- Developed parametric and linear interpolation algorithms to model household inflation density forecasts.
- Identified statistically and economically significant sensitivities of inflation forecasts to survey question design.
- Applied linear regression techniques to assess sociodemographic correlates of survey response discrepancies.

Fiscal Deficits and Inflation Risks in Advanced and Emerging Economies

with Ryan Banerjee, Aaron Mehrotra, & Fabrizio Zampolli | [BIS Working Paper #1](#), [BIS Working Paper #2](#)

- Applied quantile regression techniques to estimate the effects of fiscal deficits on inflation risk in 50 advanced and emerging economies over four decades.
- Fit cross-country inflation data to parametric distributions to assess the impact of fiscal deficits on inflation tail risks.
- Engineered a Monte Carlo simulation to quantify bias due to correlated shocks.

TECHNICAL SKILLS

Programming Languages and Tools: Python, SQL, R, Matlab, L^AT_EX, GitHub, Amazon Web Services, Redshift, ML

Methods: Differences-in-differences, regression discontinuity, instrumental variables, event study, Monte Carlo simulation, principal component analysis, linear and logistic regression, spline interpolation, hypothesis testing, classification, clustering, forecasting, natural language processing

AWARDS AND FELLOWSHIPS

Capital One Credit and Financial Risk Management, Models and Data- Groundbreakers Award 2026Q1, California Policy Lab 2021-2022 (\$27k), BB90 Fund for Monetary Economics 2023 (\$20k), Berkeley Opportunity Lab Place-Based Policy Initiative Grant 2024 (\$4k), UC Berkeley Graduate Fellowship 2018-2025

PERSONAL INFORMATION

Citizenship: USA

Languages: English (Fluent), Arabic (Fluent), French (Proficient)